

AI-Assisted Software Development

Mastering AI Coding Assistants

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The Need for Custom Instructions

Consider this scenario:

- You're working on a large project
- Your team has specific coding conventions
- You use particular libraries and frameworks
- You have performance requirements

How do you ensure the AI understands all of this? 🤔

This is where custom instructions come in!

What are Custom Instructions?

Think of custom instructions like "training" your AI assistant:

- Like teaching a new team member your project's standards
- Providing context about your codebase
- Setting rules for code generation

Benefits of Custom Instructions

Let's explore why custom instructions are valuable:

1. Customized AI Behavior

- Aligns with your project's specific needs
- More relevant code suggestions
- Fewer iterations needed

2. Consistency Within and Across Teams

- Enforces coding standards automatically
- Maintains uniform style
- Consistent AI responses for all team members

3. Enhanced Context Awareness

- AI understands your project better
- Knows about commonly used methods
- Aware of architectural decisions

4. Productivity Boost

- Less time editing AI suggestions
- Faster code generation
- Reduced back-and-forth

Specifying Custom Instructions

- Specify your custom instructions in natural language
- Tools will automatically use them and send them to the AI

Global Instructions

Cursor allows you to set global instructions:

- `Cursor Settings > General > Rules for AI`

Project-Specific Instructions

Project-specific instructions let you fine-tune the AI for your project:

Cursor AI

```
└─ your-project/  
  └─ .cursorrules
```

GitHub Copilot

```
└─ your-project/  
  └─ .github/  
    └─ copilot-instructions.md
```

Examples

Cursor Example:

```
// .cursorrules example  
Prefer using async/await over Promises  
Follow BEM naming convention for CSS  
Use TypeScript strict mode
```

GitHub Copilot Example:

```
// copilot-instructions.md  
This project:  
- Uses React 18 with TypeScript  
- Follows Atomic Design principles  
- Requires accessibility (WCAG 2.1)  
- Uses Jest for testing
```

The AI now has context for *all* interactions! 🎯

Use Cases: Coding Standards

Custom instructions can enforce:

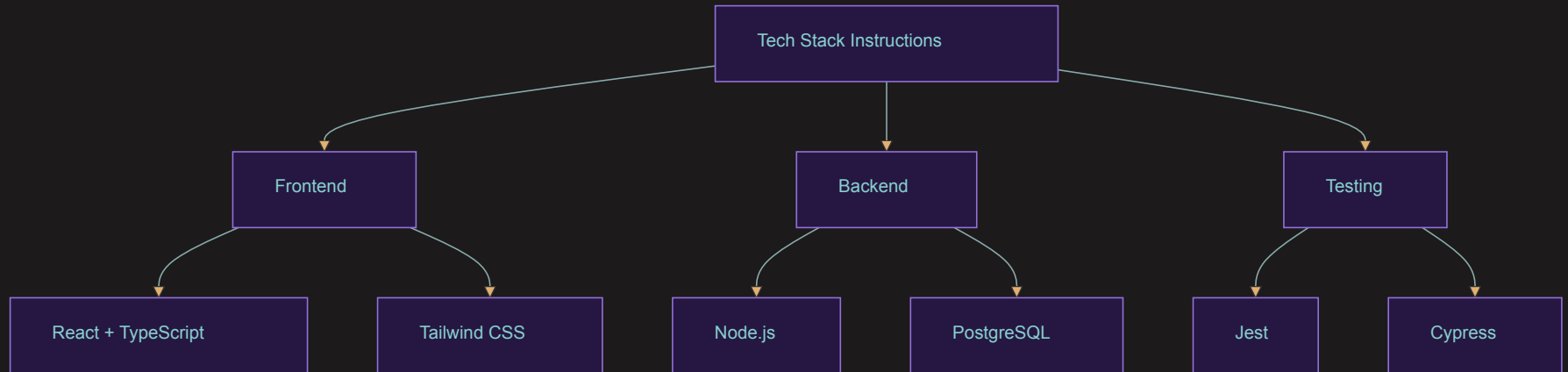
- Naming conventions

```
// ✓ Good  
interface UserProfile { ... }  
  
// ✗ Avoid  
interface userprofile { ... }
```

- Code structure
- Documentation requirements
- Testing patterns

Use Cases: Technology Stack

Guide the AI about your tech choices:



Use Cases: Performance Optimization

Tell the AI to prioritize:

- ⚡ Minimizing blocking operations
- 🔄 Implementing caching strategies
- 🎯 Code splitting
- 🚀 Resource optimization

Example:

```
// AI will prefer this pattern
const heavyOperation = async () => {
  const cached = await cache.get('key');
  if (cached)
    return cached;
  const result = await computeExpensiveResult();
  await cache.set('key', result);
  return result;
};
```

Ready-Made Instructions

GitHub repository of Cursor rules:

- <https://github.com/PatrickJS/awesome-cursorrules/>

Other directories of Cursor rules:

- <https://cursor.directory>
- <https://dotcursorrules.com>
- <https://cursorrule.com>

Summary: Maximizing AI Assistant Effectiveness

Key Takeaways:

1. Custom Instructions are Powerful

- Tailor AI behavior to your needs
- Improve code quality and consistency

2. Implementation is Simple

- `.cursorrules` for Cursor AI
- `copilot-instructions.md` for GitHub Copilot

3. Benefits are Significant

- Better code suggestions
- Increased productivity
- Improved team alignment

Remember: The better your instructions, the better your AI assistant performs! 🚀